LEARNER GUIDE

# Putting it together

Introduction to Trigonometry. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

## What this lesson is about

This lesson gathers the whole unit into one skill: choosing. You already know SOH · CAH · TOA, how to find a side, and how to find an angle. What is left is reading a problem and deciding which of those methods it needs. Every right-angled-triangle problem starts with the same two questions. First: am I finding a side or an angle? If the angle is given and a length is missing, you are finding a side; if two lengths are given and the angle is missing, you are finding an angle. Second: which ratio do the sides make? Opposite and hypotenuse give sine; adjacent and hypotenuse give cosine; opposite and adjacent give tangent. Once both are answered, the method follows — for a side: write the ratio, substitute, rearrange, evaluate; for an angle: write the ratio, substitute, apply the inverse, evaluate. You will diagnose and solve mixed problems, judge complete worked solutions for flaws, and reveal real measurements — the height a ladder reaches, the drop of a zip-wire, the pitch of a roof. The question you are exploring is: how do I choose my method?

## What you are learning to notice

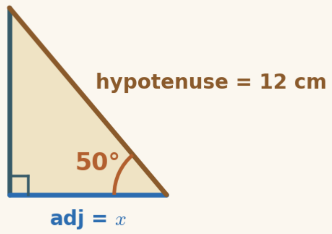
- That every right-triangle problem begins with two questions: a side or an angle? and which ratio?
- That the diagnosis chooses the method — finding a side and finding an angle are the two paths, and the clues in the problem tell you which.
- That checking finished work for flaws (wrong ratio, multiplying instead of dividing, forgetting the inverse) is how you learn to check your own.
- That the same two questions unlock real objects — ladders, ramps, roofs, zip-wires — from measurements you could actually take.

## Moving through it, one step at a time

In Side or angle?, diagnose each problem: decide side-or-angle, then the ratio, and reveal the full method. In the Trig Problem Studio, do it unaided — Step 1 a side or an angle?, Step 2 which ratio?, then work out the answer on your calculator. In Spot the flaw, read a complete worked solution and judge whether it is right or flawed. In Reveal it in the world, decide which kind each real object needs and solve it. Keep your calculator in degrees, and use the built-in ■ Calculator (with the inverse keys) and the √ Symbols palette under each answer box.

## Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

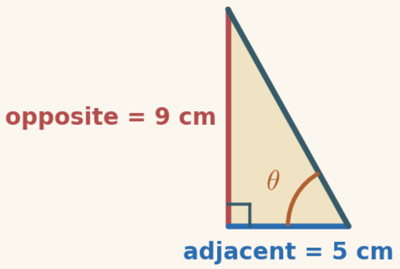



### Example 1 — diagnosing a side problem

The angle (50°) is given and the adjacent side is missing — so this is a finding-a-side problem. The adjacent and hypotenuse go together, so use cosine.

1. a side or an angle? — a side
2. which ratio? (adjacent & hypotenuse) — $\cos 50° = \frac{x}{12}$
3. rearrange ($x$ is on top, so multiply) — $x = 12 \times \cos 50°$
4. work it out — $x = 12 \times 0.643$

**$x$ = 7.71 cm**




### Example 2 — diagnosing an angle problem

This time two sides are given and the angle is missing — so this is a finding-an-angle problem. The opposite and adjacent go together, so use the tangent, then its inverse.

1. a side or an angle? — an angle
2. which ratio? (opposite & adjacent) — $\tan \theta = \frac{9}{5} = 1.8$
3. apply the inverse (undo tan) — $\theta = \tan^{-1}(1.8)$
4. work it out (degrees) — $\theta = 60.9°$

**$\theta$ = 60.9°**

### A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

## If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (SOH · CAH · TOA and choosing a ratio (Lesson 4), finding a side by writing and solving the ratio equation, including when the unknown is in the denominator (Lessons 5–6), and finding an angle with the inverse ratios (Lesson 7)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.

- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

**How the worked solution unlocks**

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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